Experiment

A1-1

Korean (Korea (Republic of))

LED의 색

Part A. 보정 (0.6점)

A.1 (0.3pt)

저장한 스펙트럼: R.csv G.csv B.csv

A.2 (0.1pt)

색	N
빨강	
초록	
파랑	

A.3 (0.2pt)

$\text{slope} =$ $\text{shift} =$

Part B. 공기 중의 LED (5.3점)

B.1 (0.1pt)

측정 범위, V	R_V
600	
200	
20	
2	
200m	

B.2 (0.1pt)

$R_0 =$ $U_{100\,\text{mA}} =$

B.3 (0.3pt)

회로도:

B.4 (0.6pt)

V_1 ( )	V_2 ( )

B.5 (0.1pt)

(1) + ○—▷|—○ –

(2) + ○—|◁—○ –

올바른 극성: (1) / (2)

B.6 (0.3pt)

회로도:

B.7 (0.9pt)

V_1 ( )	V_2 ( )

B.8 (0.4pt)

저전류 모드:

$I =$

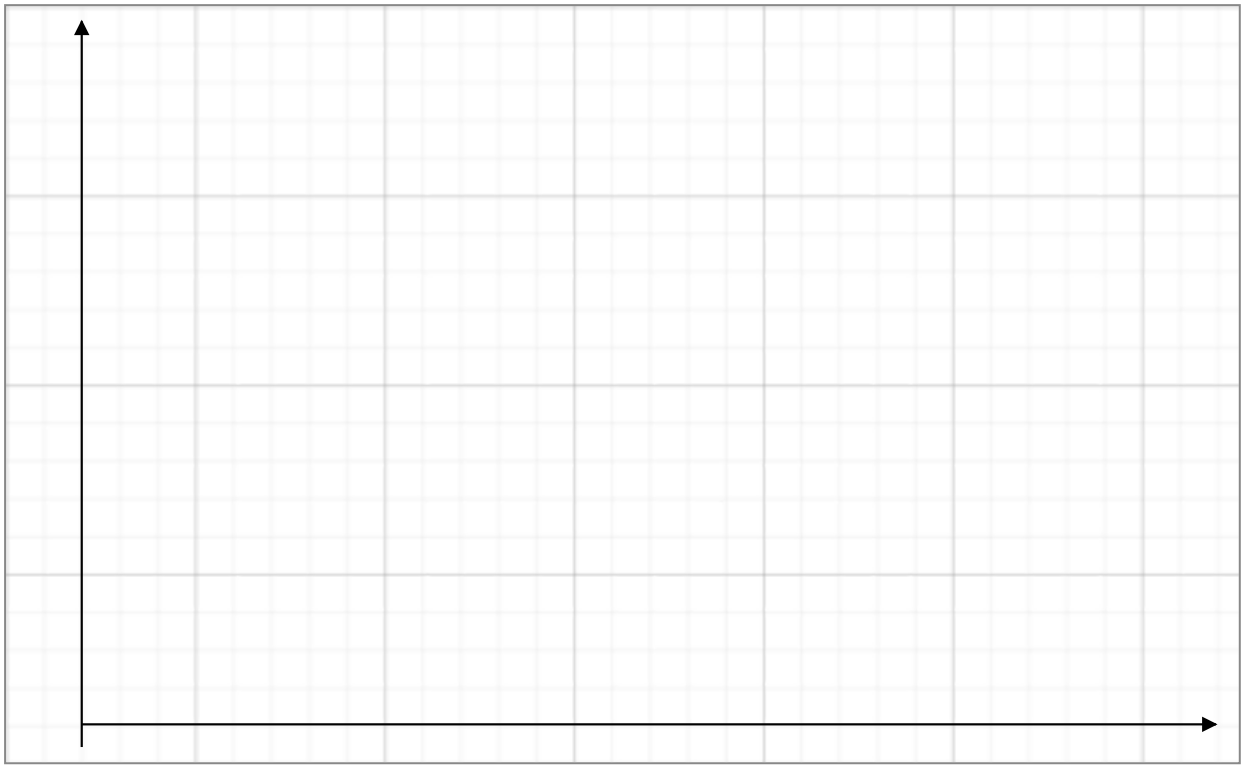
$U =$

고전류 모드:

$I =$

$U =$

B.9 (0.2pt)
그래프:



B.10 (0.1pt)

$\eta_0 k_B T_0 / e =$

B.11 (0.3pt)

$\eta_{\text{room}} =$

B.12 (0.8pt)

I, mA	λ, nm

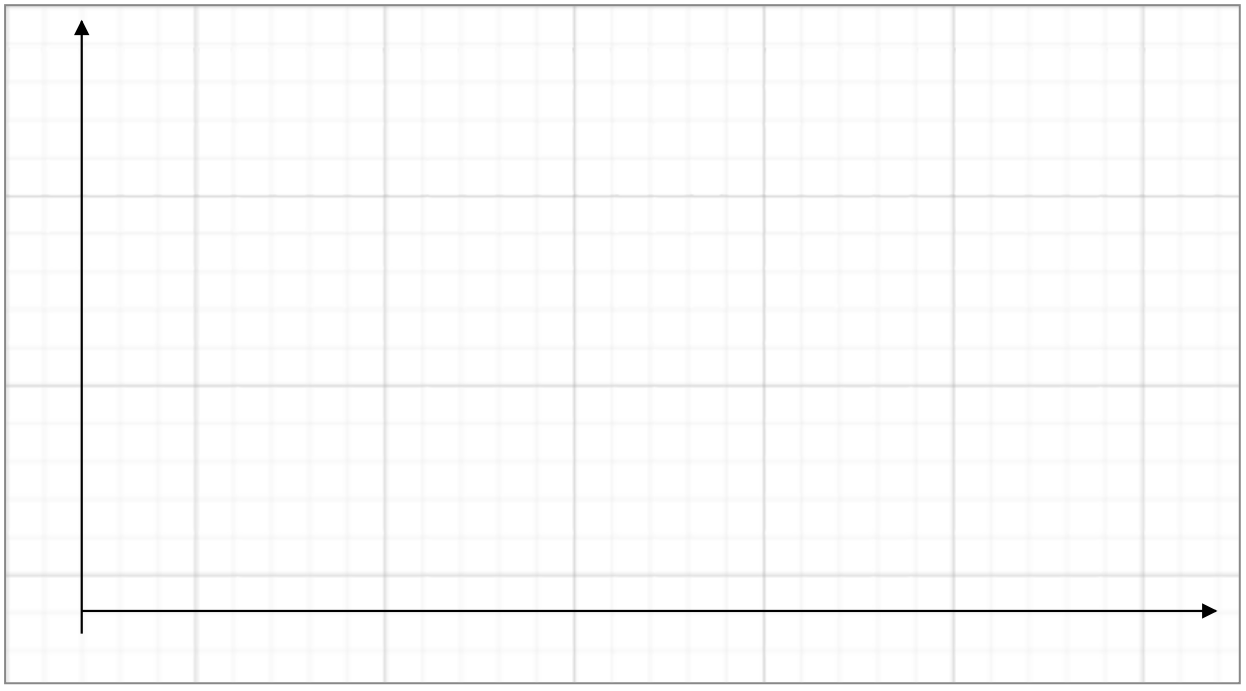
B.13 (0.8pt)

저장한 시계열:

B.14 (0.3pt)

t, s	λ, nm

그래프:



Part C. 물속의 LED (3.5점)

C.1 (0.9pt)

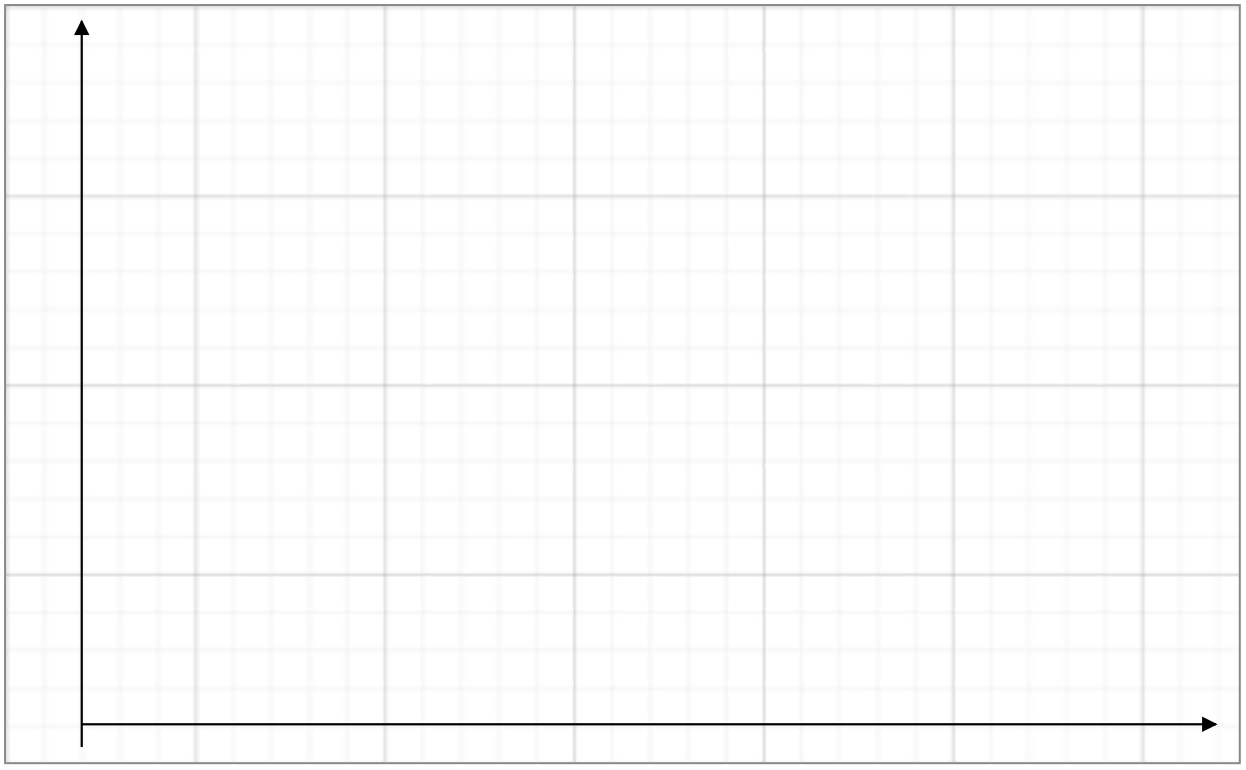
V_1 ( )	V_2 ( )

C.2 (0.2pt)

$I =$

$U =$

C.3 (0.5pt)
그래프:



C.4 (0.8pt)

I, mA	λ, nm

C.5 (0.8pt)

저장한 시계열:

C.6 (0.3pt)

t, s	λ, nm

Part D. 액체 질소 속의 LED (6.1점)

D.1 (0.1pt)

저장한 스펙트럼:

D.2 (0.6pt)

V_1 ( )	V_2 ( )

D.3 (0.9pt)

V_1 ( )	V_2 ( )

D.4 (0.2pt)

저전류 모드:

$I =$

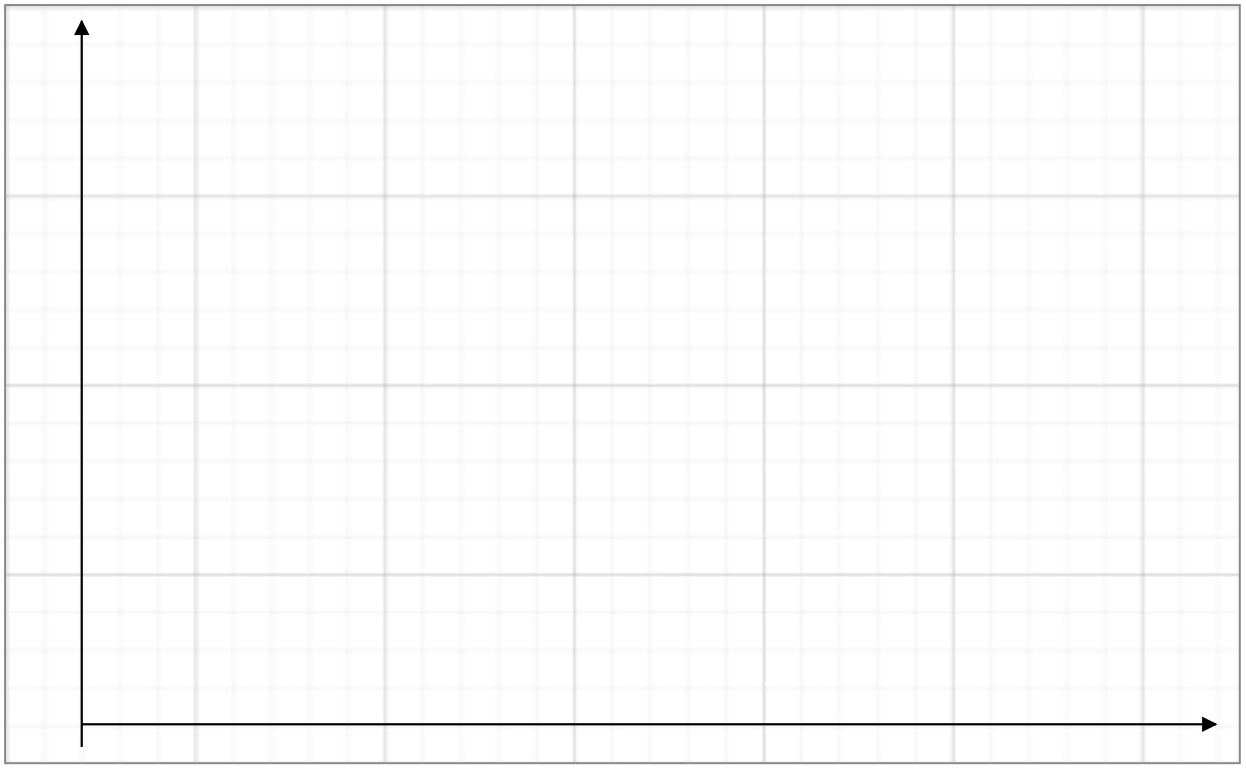
$U =$

고전류 모드:

$I =$

$U =$

D.5 (0.2pt)
그래프:



D.6 (0.3pt)

$\eta_{N_2} =$

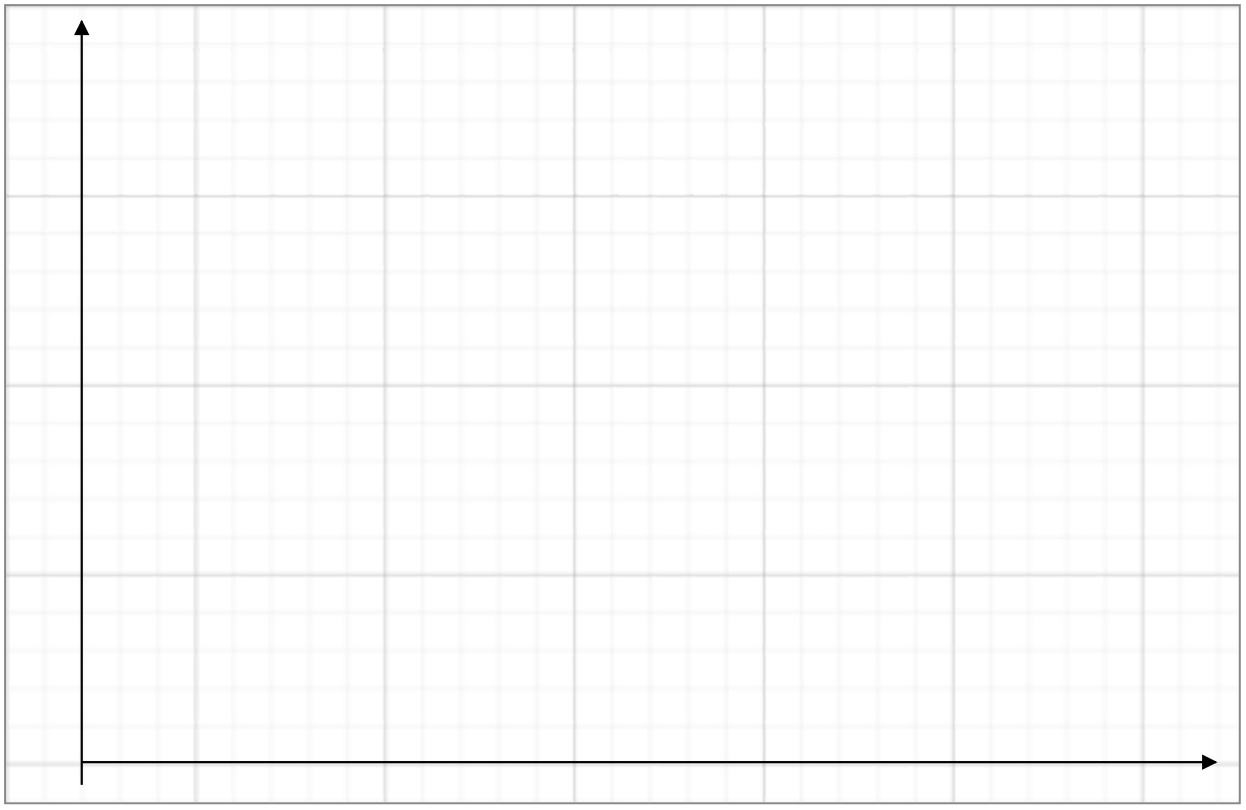
D.7 (1.0pt)

I, mA	스펙트럼 파일

D.8 (0.5pt)

I, mA	λ, nm

D.9 (1.0pt)
그래프:



D.10 (0.6pt)

$\lambda_{N_2} =$ $\lambda_{\text{room}} =$

D.11 (0.7pt)

$E_0 =$ $\alpha =$

Part E. 열량 측정 매개변수 계산 (3.1점)

E.1 (0.5pt)

(식) $\chi =$ (값) $\chi =$

E.2 (0.5pt)

$\varkappa_{\text{air}}/\varkappa_{\text{water}} =$

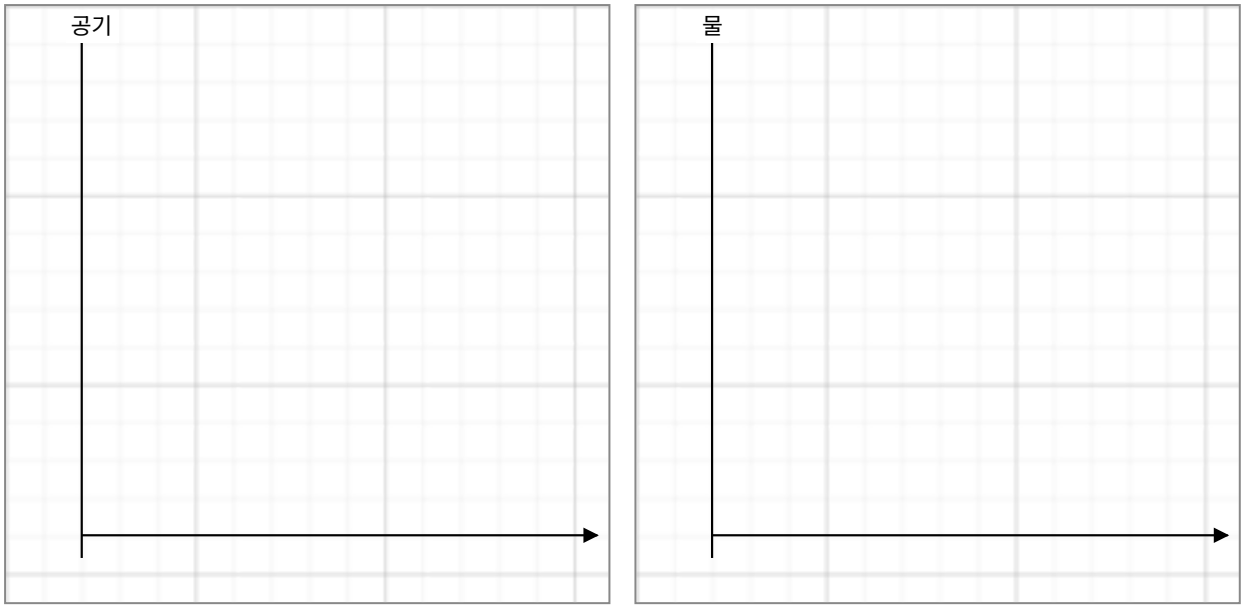
E.3 (0.3pt)

$T(t) =$

E.4 (1.0pt)

t, s (공기)	T	t, s (물)	T

E.5 (0.5pt)
그래프:



E.6 (0.3pt)

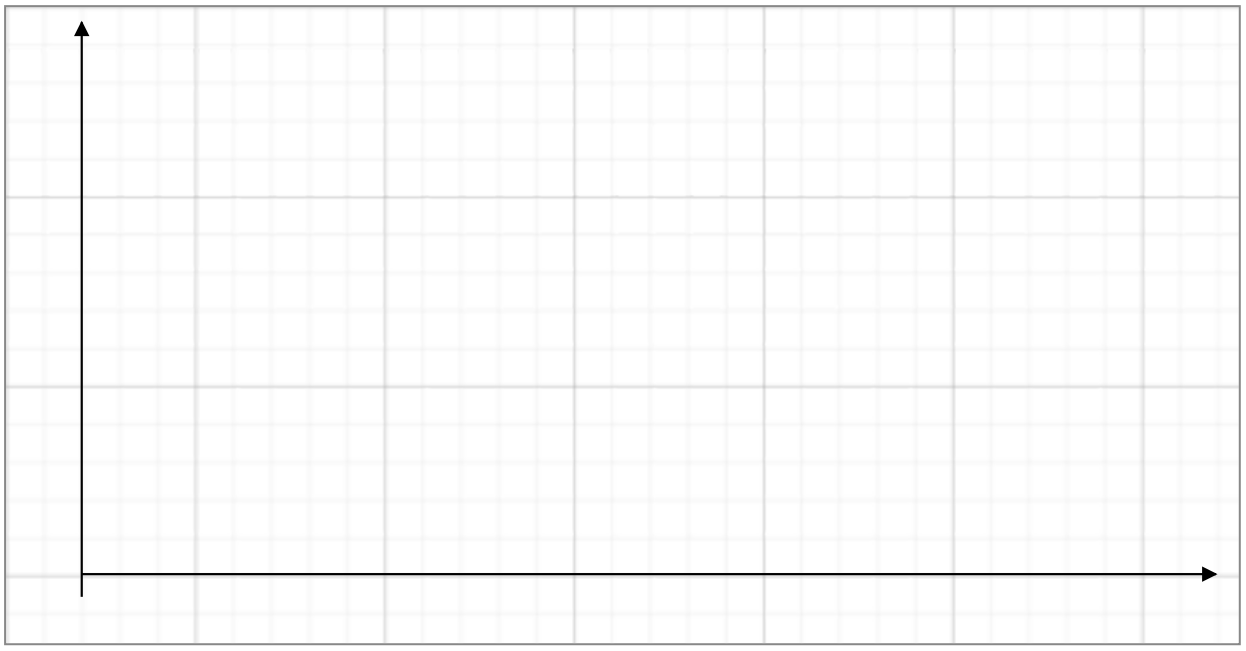
$C_{\text{air}}/C_{\text{water}} =$

Part F. 포화 전류의 온도 의존성 (1.4점)

F.1 (0.6pt)

T	$\ln(I_0/\mathrm{mA})$

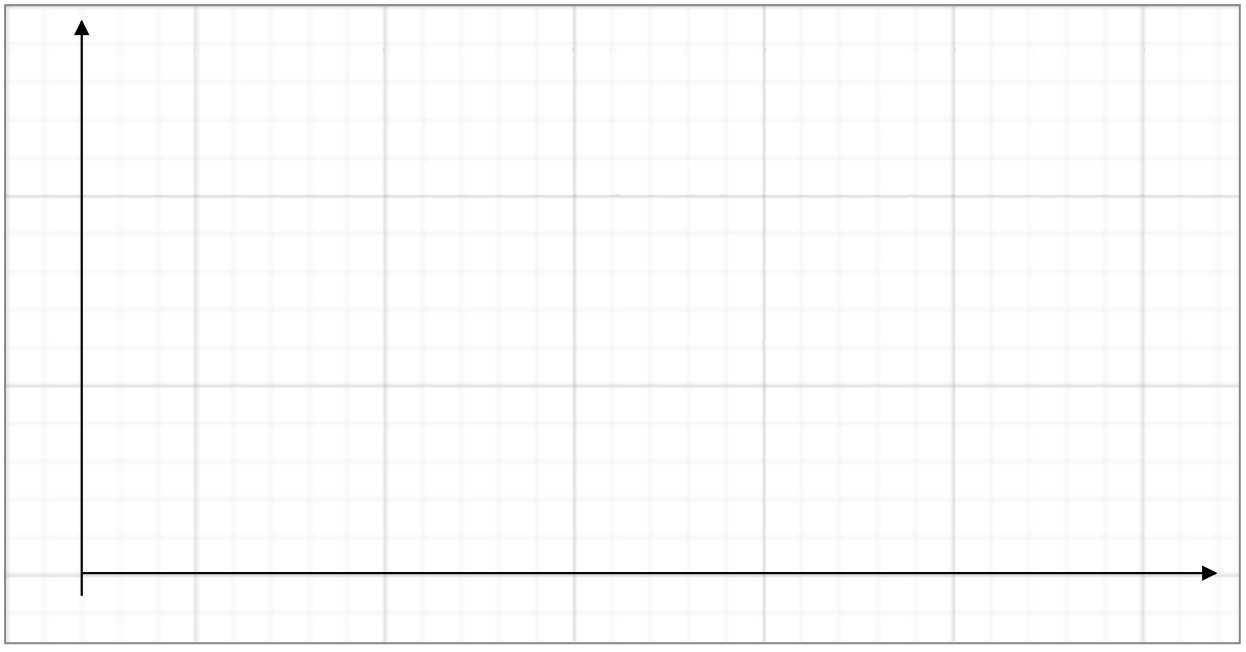
그래프:



F.2 (0.6pt)

T	$\ln(I_0/\mathrm{mA})$

그래프:



F.3 (0.2pt)

배수: